Fast Fourier Transform (FFT)

FFT for Real Valued Signal

Ira A. Fulton Schools of Engineering
Arizona State University

FFT for Real Valued Signal

- **x[n]: N-point real-valued signal.**
- **We can form a N/2-point complex signal z[n]:**

$$z[n] = x_e[n] + jx_o[n]$$

where $x_e[n] = x[2n]$, $x_o[n] = x[2n+1]$, $n = 0, 1, \ldots, N/2-1$.

- **Compute DFT X[k] using the following steps:**
  - Compute N/2-point FFT of z[n] to get Z[k], k = 0, 1, …, N/2-1.
  - Extract $X_e[k]$ and $X_o[k]$ from Z[k].
  - X[k] can be computed from $X_e[k]$ and $X_o[k]$ using butterfly structure (last stage of N-point FFT).

Conjugation Property of DFT

- **x[n]: N-point real-valued signal.**
- $x[n] \xleftrightarrow{N-point\ DFT} X[k]$
- **Complex conjugation property:**
  - $X[k] = X^*[(-k)_N]$
  - Rewrite as: $X[k] = X^*[N-k]$
  - Replace k with k + N/2: $X\left[k + \frac{N}{2}\right] = X^*\left[\frac{N}{2} - k\right]$
- **i.e., X[k] is conjugate symmetric with respect to N/2.**

How to Extract $X_e[k]$ and $X_o[k]$ from $Z[k]$?

Given:

- $z[n] = x_e[n] + jx_o[n]$ with $x_e[n] = x[2n]$ and $x_o[n] = x[2n+1]$, $n = 0, 1, \ldots, N/2-1$

Take N/2-point DFT:

- $Z[k] = X_e[k] + jX_o[k]$ (Eq.1)

Take complex conjugate:

- $Z^*[k] = X_e^*[k] - jX_o^*[k]$

Replace k with –k and take modulo N/2:

- $Z^*[(-k)_{N/2}] = X_e^*[(-k)_{N/2}] - jX_o^*[(-k)_{N/2}]$

$x_e[n]$ and $x_o[n]$ are N/2-point real-valued, by complex conjugation property:

- $X_e[k]=X_e^*[(-k)_{N/2}]$, $X_o[k]=X_o^*[(-k)_{N/2}]$

We get

- $Z^*[(-k)_{N/2}] = X_e[k] - jX_o[k]$ (Eq.2)

How to Extract $X_e[k]$ and $X_o[k]$ from $Z[k]$?

From

- $Z[k] = X_e[k] + jX_o[k]$ (Eq.1)
- $Z^*[(-k)_{N/2}] = X_e[k] - jX_o[k]$ (Eq.2)

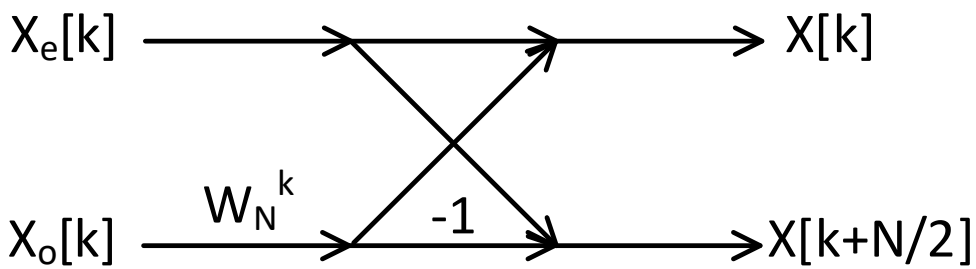
We get

- (Eq.1) + (Eq.2): $X_e[k] = (Z[k] + Z^*[(-k)_{N/2}]) / 2$
- (Eq.1) - (Eq.2): $X_o[k] = (Z[k] - Z^*[(-k)_{N/2}]) / (2j)$

How to Compute X[k] from $X_e[k]$ and $X_o[k]$?

Last stage of FFT butterfly structure (N/2 butterflies):

- $X[k] = X_e[k] + W_N^k\, X_o[k], k = 0, 1, \dots, \frac{N}{2} - 1$
- $X\left[k + \frac{N}{2}\right] = X_e[k] - W_N^k\, X_o[k], k = 0, 1, \dots, \frac{N}{2} - 1$
- $W_N^k = e^{-j\frac{2\pi}{N}k}$



Example

- **Sketch the flow graph to compute a 4-point DFT using a complex 2-point FFT for x[n] = [1, 2, 0, 0].**

Answer

- $x[n] = [1, 2, 0, 0]$
- Extract even and odd subsequences:

$$x_e[n] = x[2n] = [1, 0]$$

$$x_o[n] = x[2n+1] = [2, 0]$$

- Form a new complex-valued N/2-point signal:

$$z[n] = x_e[n] + jx_o[n] = [1+2j, 0]$$

Answer

z[n] = [1+2j, 0]

Given:

$X_e[k] = (Z[k] + Z^*[(-k)_{N/2}]) /2$

$X_o[k] = (Z[k] - Z^*[(-k)_{N/2}]) / (2j)$

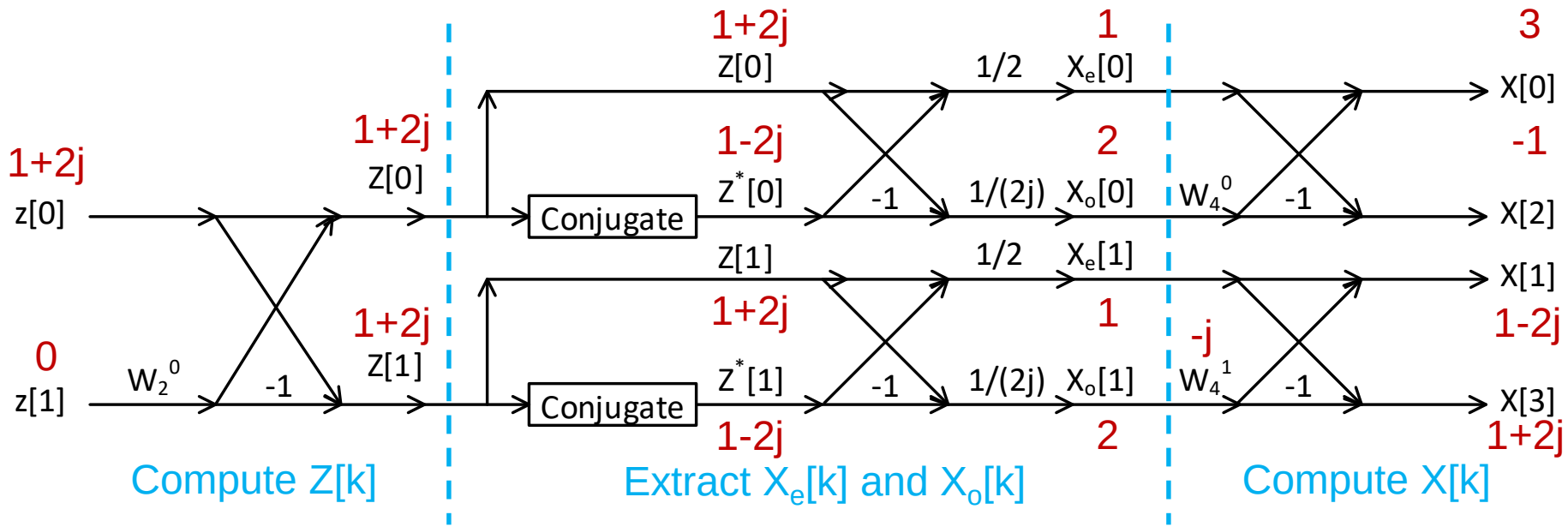
We can extract

$X_e[0] = (Z[0] + Z^*[(-0)_2]) / 2 = (Z[0] + Z^*[0]) / 2$

$X_e[1] = (Z[1] + Z^*[(-1)_2]) / 2 = (Z[1] + Z^*[1]) / 2$

$X_o[0] = (Z[0] - Z^*[(-0)_2]) / (2j) = (Z[0] - Z^*[0]) / (2j)$

$X_o[1] = (Z[1] - Z^*[(-1)_2]) / (2j) = (Z[1] - Z^*[1]) / (2j)$



X[k] = [3, 1-2j, -1, 1+2j]

Computation Complexity

- **For N/2-point FFT:**
  - $\log_2(N/2)$ stages
  - N/4 butterflies per stage
  - Total FFT butterflies: $\log_2(N/2) * N/4$
- **N/2 butterflies to extract $X_e[k]$ and $X_o[k]$ from Z[k]**
- **N/2 FFT butterflies to compute X[k] from $X_e[k]$ and $X_o[k]$**
- **Total number of FFT butterflies: $\log_2(N/2) * N/4 + N/2$**
- **Total computations (simplified): $(\log_2(N/2)*N + 2N)$ MULTs**
  - 1 CMULT = 4 MULTs + 2 ADDs per butterfly
- **Compared to N-point FFT: $2N*\log_2(N)$ MULTs**

Exercise

- Sketch the flow graph to compute a 4-point DFT using a complex 2-point FFT for x[n] = [2, 1, -1, 3].
- Determine the number of (real) multiplications needed to compute a 4-point DFT using a complex 2-point FFT.
- Determine the number of (real) multiplications needed to compute a 4-point DFT using a 4-point FFT.
- Determine the number of (real) multiplications needed to directly compute a 4-point DFT.

Answer

- $x[n] = [2, 1, -1, 3]$
- $z[n] = [2+j, -1+3j]$

Given:

$X_e[k] = (Z[k] + Z^*[(-k)_{N/2}])\ /2$

$X_o[k] = (Z[k] - Z^*[(-k)_{N/2}])\ /\ (2j)$

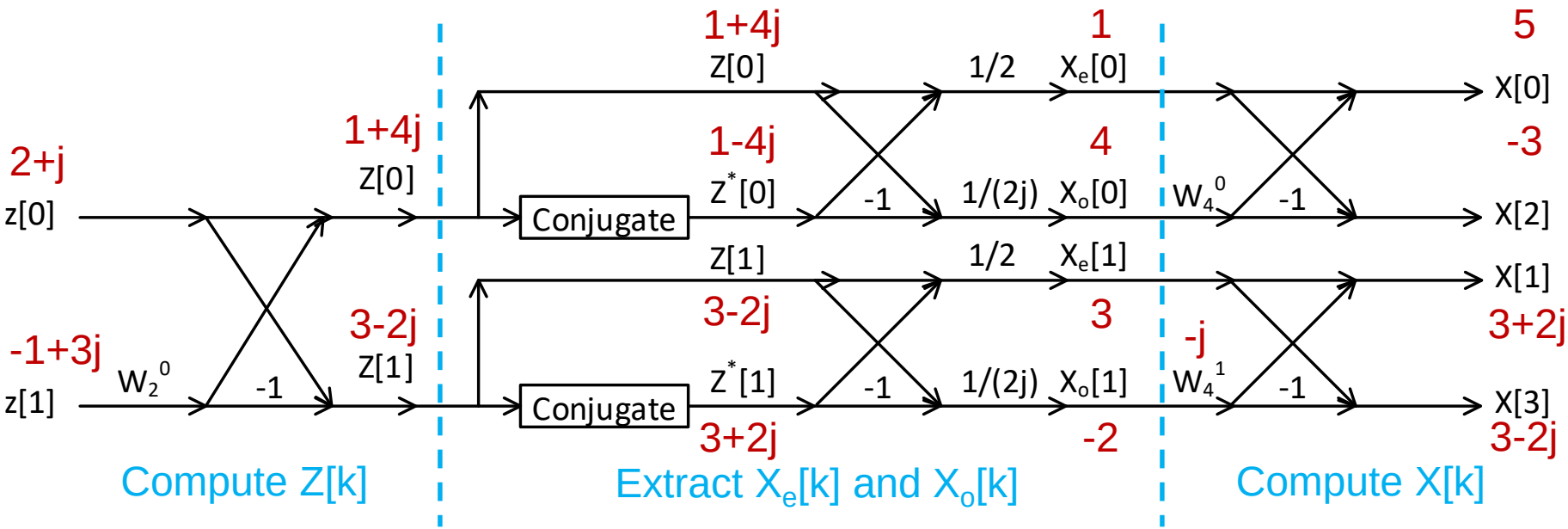
We can extract

$X_e[0] = (Z[0] + Z^*[(-0)_2])\ /\ 2 = (Z[0] + Z^*[0])\ /\ 2$

$X_e[1] = (Z[1] + Z^*[(-1)_2])\ /\ 2 = (Z[1] + Z^*[1])\ /\ 2$

$X_o[0] = (Z[0] - Z^*[(-0)_2])\ /\ (2j) = (Z[0] - Z^*[0])\ /\ (2j)$

$X_o[1] = (Z[1] - Z^*[(-1)_2])\ /\ (2j) = (Z[1] - Z^*[1])\ /\ (2j)$



- $X[k] = [5, 3+2j, -3, 3-2j]$

Answer

- **The number of (real) multiplications needed to compute a 4-point DFT using a complex 2-point FFT:**
  - $(\log_2(N/2)*N + 2N) = \log_2 2 * 4 + 8 = 12$ MULTs
- **The number of (real) multiplications needed to compute a 4-point DFT using a 4-point FFT:**
  - $2N*\log_2(N) = 2 * 4 * \log_2 4 = 16$ MULTs
- **Determine the number of (real) multiplications needed to directly compute a 4-point DFT:**
  - $4N^2 = 4 * 4^2 = 4 * 16 = 64$ MULTs

Question

For a N = 64-point DFT, how many real multiplication operations are needed if it is implemented using a complex 32-point FFT?

a) 448

b) 768

c) 4096

d) 16384

FFT for Two Real Valued Signals

- **$x_1[n]$ and $x_2[n]$ are two N-point real valued signals.**
- **Form one new complex-valued N-point signal:**
  - $z[n] = x_1[n] + jx_2[n]$
- **Compute N-point FFT of z[n] to get Z[k].**
- **Extract $X_1[k]$ and $X_2[k]$ from Z[k] and stop.**

Recall: Frequency Resolution of DFT

- **DFT:** $X[k] = X\left(e^{j\omega}\right)\Big|_{\omega_k = k\frac{2\pi}{N}}$
- **Digital frequency resolution in radians:** $\Delta\omega = \frac{2\pi}{N}$
- **Analog frequency resolution in radians/second:** $\Delta\Omega = \frac{2\pi}{N} f_s$
- **Analog frequency resolution in Hz:** $\Delta f = \frac{f_s}{N}$
- **As N increases, frequency resolution improves, but computation complexity also increases.**

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